Jayadeep Gowda K B

Software Engineering | AI & Intelligent Systems | IoT & Automation

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Education

Bangalore Institute of Technology, Bangalore 2027
Bachelor of Engineering in Robotics & Artificial Intelligence | *CGPA:* ***8.9***

JS Pre-University College, Bangalore 2023
Class XII, KSEAB | *Percentage:* ***96***

St. Philomena's High School, Bangalore 2021
Class X, KSEEB | *Percentage:* ***94***

Technical Skills

- **Languages:** Python, C++, JavaScript, SQL
- **AI/ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, Transformers, CNNs, LLMs, OpenCV, YOLO
- **IoT, Edge AI & Robotics:** Raspberry Pi, ARM Deployment, ROS2, ZeroMQ, Sensor Fusion, Edge Inference
- **Agentic AI & LLM Frameworks:** LangChain, LangGraph, RAG Pipelines, Multi-Agent Workflows
- **Cloud, DevOps & Data:** AWS, Docker, CI/CD, Git, Linux, PostgreSQL, PostGIS, MongoDB, ChromaDB
- **Software Engineering:** REST APIs, OOP's, FSM/State-Machine Architecture, React, Angular, FastAPI, Pydantic

Experience

Indian Institute of Science (IISc) March – June 2026
Research Intern – Dept. of Computational & Data Sciences

- Engineered a real-time **sensor data pipeline** ingesting 5 concurrent Crystal-MXene streams at 5.68–26.32 Hz; unified disparate sampling rates into a single analytics framework.
- Benchmarked **17 ML/DL architectures** across 21 time/frequency-domain features with 5-fold stratified CV; delivered production-grade, unit-tested Python codebase.
- Drove **cross-functional collaboration** across 2 countries — iterated models through 10+ code review cycles with Postdocs, PhDs, and partners at AGH University, Poland.

Sirena AI Systems January – Feb 2026
R&D Intern – Automation & Edge AI

- Owned **end-to-end development** of a production autonomous robot — from PyBullet simulation to embedded ARM deployment — across 3 morphologies (Humanoid, Quadruped, Hexapod).
- Architected **12-state FSM controllers** and custom inverse-kinematics solvers in Python/C++; applied reusable OOP design patterns, reducing per-morphology boilerplate by ∼40%.
- Built a real-time **perception stack** (YOLOv8 + Whisper + on-device LLM) on ARM-based IoT nodes; achieved **¡50 ms** latency via ZeroMQ pub-sub messaging.

Frillory Design House June – Aug 2025
AI Automation Engineer

- Architected an **agentic LLM automation pipeline** orchestrating prompt chains and generative models end-to-end; automated 50+ enterprise design assets, cutting manual overhead by **25%**.

Projects & Publication

MEMO (Neural-OS) – Multimodal Autonomous IoT Robot Agent *[Paper Accepted]*

- Designed and shipped an **autonomous multimodal IoT assistant** on Raspberry Pi 5 — engineering event-driven pipelines for vision (YOLOv8, FaceNet), speech (Whisper), and local LLM inference.
- Optimized a 2.7B-param LLM via **4-bit quantization** to 10–14 tok/s on edge hardware, achieving **600 ms end-to-end latency**; won **Best Poster Award** for technical innovation.
- Research paper on real-time sensor fusion and autonomous decision-making on resource-constrained IoT devices **accepted for publication**, 2026.

RILA – AI Agent for Searchable Video Memory

- Built an end-to-end **AI agent** converting YouTube/Instagram reels into structured, searchable knowledge graphs via Whisper ASR and LangChain-orchestrated multi-step LLM extraction.
- Engineered a hybrid **RAG retrieval backend** (PostgreSQL/PostGIS + ChromaDB) enabling geospatial and semantic vector search, exposed via a REST API and Telegram bot.
- Deployed as **Dockerized microservices** with Pydantic schema validation, geocoding, and fuzzy entity matching — improving data quality and system reliability.

Leadership & Recognition

- **Competitions:** 2nd Place at GlitchVerse 2k25 Project Expo | Best Poster Award for MEMO | National Finalist, Anveshana 2025 Engineering Competition.
- **Leadership:** Joint Secretary, ROBO CELL Club – coordinated club heads and managed cross-functional workflows; Core Organizer, Ideathon 2025 (600+ participants)